YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 1

CANDIDATE NAME	

CG		INDEX	

BIOLOGY — 8876/01

Paper 1 Multiple Choice — **17 Sep 2021**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid/tape.
Write your name, index no. and CG on this cover page and Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **15** printed pages and **1** blank page.

1 A particular genetic disorder causes the progressive degeneration of nerve cells, eventually leading to death. It is caused by a defective enzyme associated with one of the cell's organelles. The enzyme is normally responsible for the breakdown of glycolipids in the cell surface membrane. In the absence of the functional enzyme, accumulation of lipids within the nerve cells leads to the death of these cells.

Which organelle is the defective enzyme likely to be associated with?

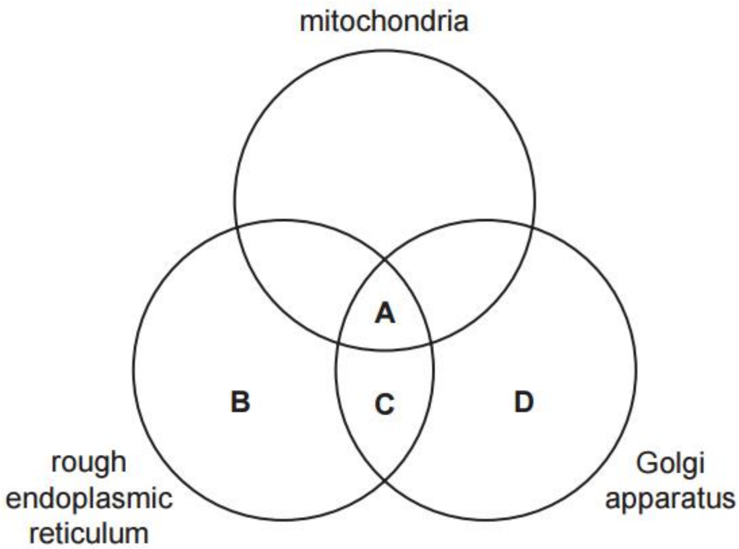
A lysosome

B nucleus

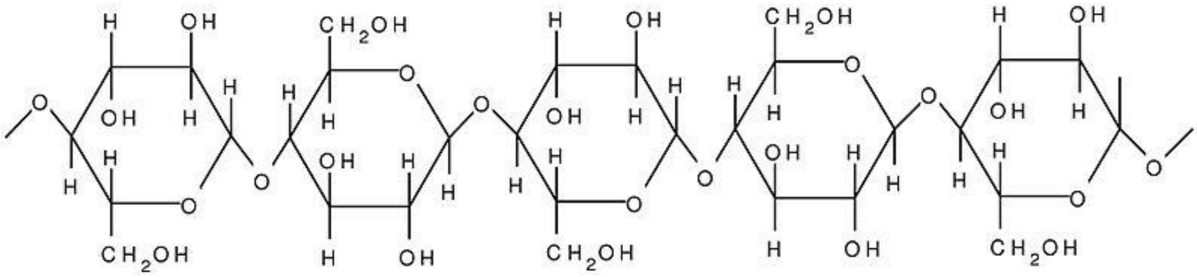
C rough endoplasmic reticulum

D smooth endoplasmic reticulum

2 Which of the following organelle(s) is/are directly required for the formation of the hydrolytic enzymes found in lysosomes?



3 The figure shows a portion of a polymer.



Which statement is **true** about the polymer?

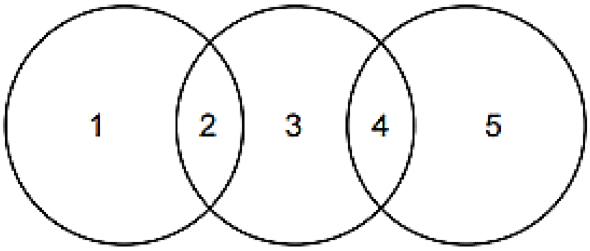
A The polymer will assume a helical structure.

B The polymer can exist in both branched and unbranched forms.

C The orientation of the monomer will result in a straight chain polymer.

D The monomers are able to form α (1-6) glycosidic bonds with one another.

4 The diagram shows the relationship between the levels of protein structure and bonds.



Which row is correct?

	1	2	3	4	5
A	primary	peptide	secondary	ionic	tertiary
B	secondary	hydrogen	tertiary	ionic	quaternary
C	tertiary	ionic	primary	peptide	quaternary
D	quaternary	ionic	tertiary	ionic	secondary

5 The statements are about the loading of haemoglobin with oxygen.

1 As the concentration of oxygen increases, the loading of oxygen increases.

2 Haemoglobin that has not bound to oxygen has a higher affinity for 2,3-bisphosphoglycerate (2,3-BPG) than for oxygen. 2,3-BPG is present in all red blood cells.

3 Binding of some oxygen to haemoglobin increase the affinity of haemoglobin for additional oxygen.

4 Binding of oxygen to haemoglobin causes any bound carbon dioxide or bound 2,3-BPG to be expelled.

5 Low pH reduces the affinity of haemoglobin for oxygen.

What explains the effect of a low concentration of oxygen on the loading of oxygen by haemoglobin?

A The affinity of haemoglobin for oxygen decreases at low concentrations of oxygen since carbon dioxide replaces oxygen, causing the pH to fall.

B Carbon dioxide remains bound to amino acids in the haemoglobin molecule and therefore makes these sites unavailable for the binding of oxygen.

C At low oxygen concentrations, oxygen is unable to compete with 2,3-BPG for binding sites with haemoglobin.

D At low oxygen concentrations, carbon dioxide expels 2,3-BPG from haemoglobin, decreasing the affinity of haemoglobin for oxygen.

6 The following are all observations about cell surface membranes.

1 Phospholipids with radioactive phosphate groups change position over time.

2 Proteins labelled with fluorescent dyes change position over time.

3 Lectins are proteins that bind to polysaccharides. Lectins only attach to the outside of cell surface membranes.

4 Electron microscope images of membranes fractured by freezing show a large number of regularly arranged particles interrupted by large particles.

5 Some proteins can easily be removed from the membrane by changing the ionic balance.

6 Some proteins can only be separated from the membrane by disrupting the membrane with detergent.

Which observations provide evidence for the fluid mosaic model of cell surface membranes?

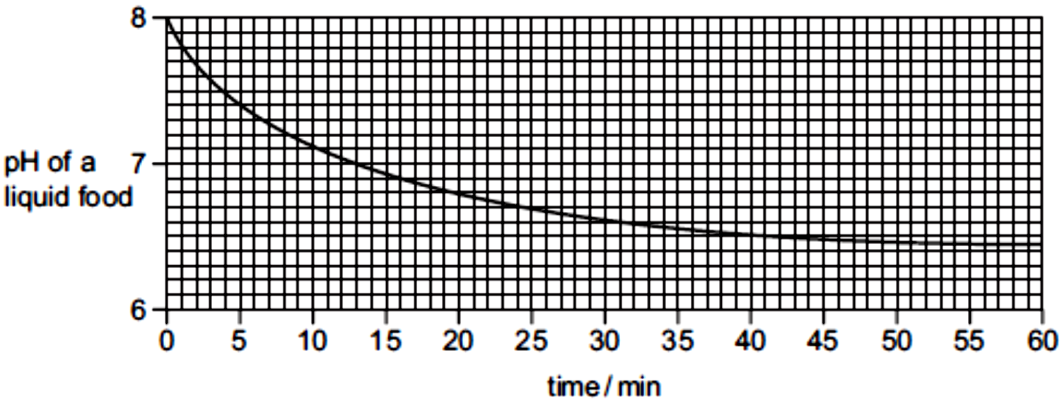
A 1, 2 and 4

B 1, 2 and 6

C 3, 4 and 5

D 3, 5 and 6

7 Lipase is a digestive enzyme produced by the pancreas that catalyses the hydrolysis of dietary lipids. The table shows how the pH of a liquid food containing a high proportion of lipids decreases over time.



Which of the following statements are possible explanations of the results of the experiment between 50 and 60 minutes?

1 Enzyme concentration becomes the limiting factor.

2 Substrate concentration becomes the limiting factor.

3 All the enzyme active sites are saturated.

4 Denaturation of the enzyme by the products.

5 Products are acting as inhibitors.

A 1, 2 and 3

B 1, 4 and 5

C 2, 3 and 4

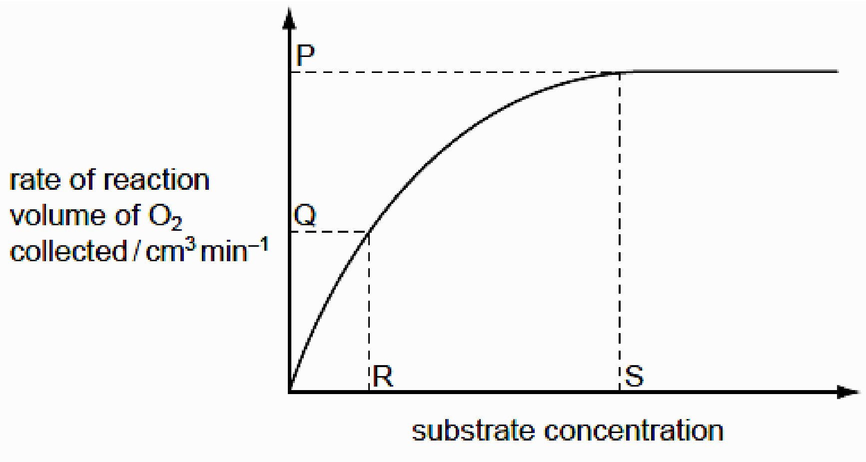
D 2, 4 and 5

8 Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time.

Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are correct?

1 At P, the rate of reaction is limited by the concentration of enzyme.

2 At Q, all of the enzyme active sites are occupied by substrate molecules.

3 At Q, the rate of reaction is limited by the concentration of the substrate.

4 At S, all of the enzyme active sites are occupied by substrate molecules.

A 1 and 4 only

B 2 and 3 only

C 2 and 4 only

D 1, 3 and 4 only

9 In a multicellular organism, which of these statements about mitosis can help to explain the control of the mitotic cell cycle?

1 In most cells the genes initiating mitosis are not switched on.

2 Mitosis produces cells to replace damaged cells that cannot be repaired.

3 Mitosis transmits a complete copy of all the alleles in a cell to new cells.

4 Daughter cells formed by asexual reproduction develop from unspecialised cells.

A 1 only

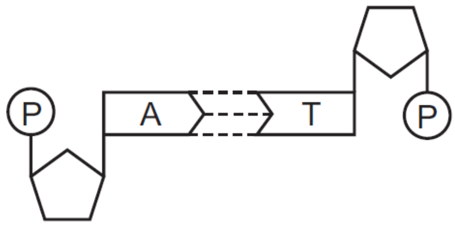
B 3 only

C 1 and 4

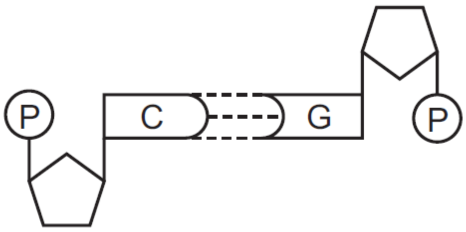
D 2 and 4

10 Which diagram represents a correct base pair of DNA?

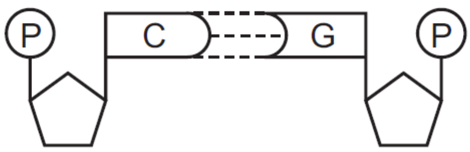
A



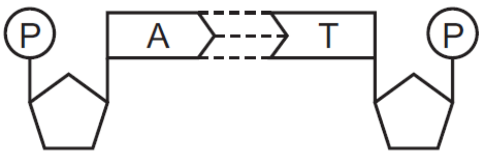
B



C

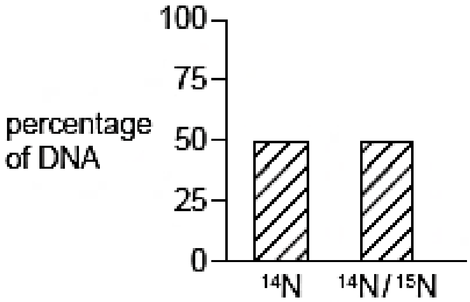


D



11 Bacteria were grown in a medium containing ^{15}N. After several generations, all of the DNA contained ^{15}N. Some of these bacteria were transferred to a medium containing the common isotope of nitrogen, ^{14}N. The bacteria were allowed to divide once. The DNA of some of these bacteria was extracted and analysed. This DNA was all hybrid DNA containing equal amounts of ^{14}N and ^{15}N.

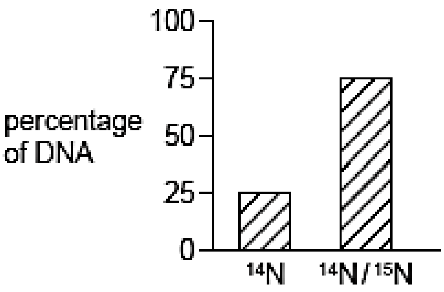
Some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N. The bacteria were allowed to divide twice. The graph shows the percentages of ^{14}N and ^{15}N in the DNA of these bacteria.



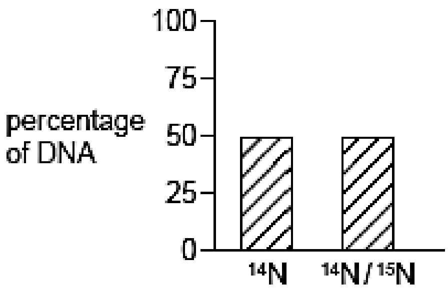
Some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N. The bacteria were allowed to divide three times.

What would be the percentages of ^{14}N and ^{15}N in the DNA extracted from these bacteria?

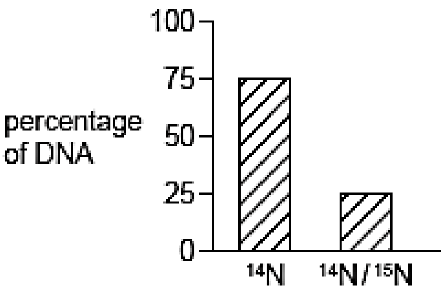
A



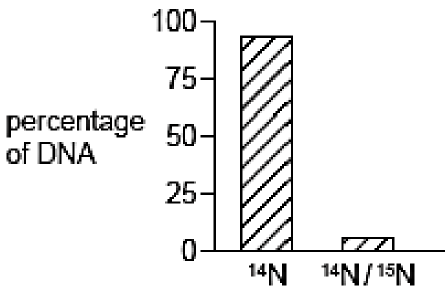
B



C



D



12 Scientists have modified the cellular machinery of *Escherichia coli* to allow synthesis of proteins from mRNA codons consisting of four nucleotides, rather than three. By significantly increasing the number of different codons, this provides the potential to synthesise novel proteins that incorporate synthetic amino acids.

For example, the four nucleotide codon AGGA was assigned to code for the synthetic amino acid p-azido-L-phenylalanine.

Which parts of the cellular machinery of *E. coli* need to be modified before synthesis of these novel proteins can occur?

A mRNA and RNA polymerase

B ribosomes and tRNA

C RNA polymerase and ribosomes

D tRNA and nucleotides

13 Which statements about the genetic code are correct?

1 Codons act as 'start' and 'stop' signals during transcription.

2 The genetic code is degenerate.

3 Prokaryotes generally use the same genetic code as eukaryotes.

4 There is only one codon for the amino acid methionine.

A 1 and 2

B 1, 2 and 3

C 1, 2 and 4

D 2, 3 and 4

14 The following sequence of bases shows a short section of linear DNA from which mRNA is transcribed.

TACTCACATTAG…

The table shows a number of mRNA codons and their corresponding amino acids.

codon	AGU	AUC	AUG	CAU	GUA	UAC	UAG	UCA
amino acid	serine	isoleucine	methionine	histidine	valine	tyrosine	'stop'	serine

Which row shows how this short section of linear DNA would be translated into part of a polypeptide chain?

	tRNA anti-codon order	amino acid sequence
A	AUGAGUGUAAUC	methionine, serine, valine, isoleucine
B	AUGAGUGUAAUC	tyrosine, serine, histidine, stop
C	UACUCACAUUAG	methionine, serine, valine, isoleucine
D	UACUCACAUUAG	tyrosine, serine, histidine, stop

15 In most organisms, six different triplets of the DNA strand that is complementary to mRNA code for the amino acid serine: AGA, AGG, AGT, AGC, TCA and TCG.

In the yeast *Candida albicans*, a seventh DNA triplet, GAC, also codes for serine. In most organisms, this triplet codes for leucine.

The diagram shows part of an mRNA molecule from *C. albicans*.

	– AGU –	UCG –	CGG –	UCA –	AGC –	ACC –	UGG –
codon number	– 11 –	12 –	13 –	14 –	15 –	16 –	17 –

Which mutation of the DNA that is complementary to this mRNA could result in *C. albicans* producing a polypeptide with an unbroken sequence of five serines in it?

A substituting a purine with a pyrimidine in the DNA coding for codon 13

B substituting a purine with a pyrimidine in the DNA coding for codon 16

C substituting a pyrimidine with a purine in the DNA coding for codon 13

D substituting a pyrimidine with a purine in the DNA coding for codon 16

16 The locus of the gene refers to the

A chromosome where the gene is found.

B position of the gene on a specific chromosome.

C position of the gene on the X chromosome.

D type of allele present on the chromosome.

17 A cell in the G1 phase has two homologous pairs of chromosomes. It then undergoes two mitotic divisions.

At the end of the second mitotic division, what is the total number of chromosomes and gene loci found in all the daughter cells formed?

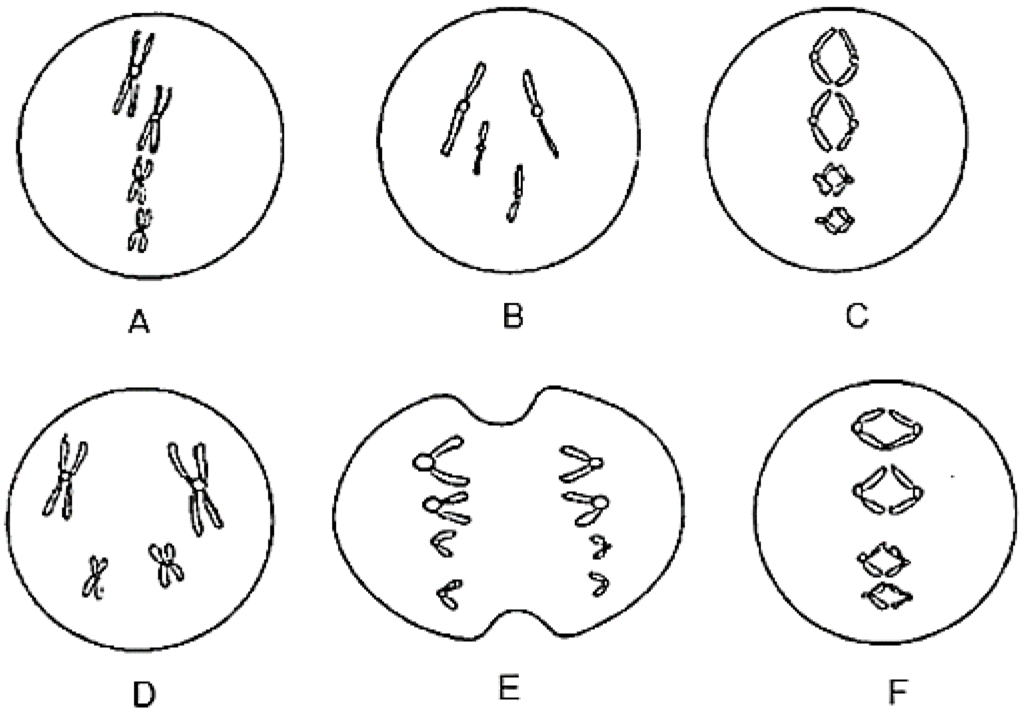
A 8 chromosomes and 4 times as many gene loci as the original parent cell.

B 8 chromosomes and 8 times as many gene loci as the original parent cell.

C 16 chromosomes and 4 times as many gene loci as the original parent cell.

D 16 chromosomes and 8 times as many gene loci as the original parent cell.

18 The diagrams below represent the chromosomes during stages in the process of mitosis.



What is the correct sequence in which these stages occur?

A B D A C F E

B B D A F C E

C D A F C E B

D D B C F E A

19 What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversion of segments of chromosomes does occur.
C	Both affect DNA sequence.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

20 Which of the following is not a characteristic of cancer cells?

A Anchorage independence

B Contact inhibition

C Metastasis

D Stimulate angiogenesis

21 Black fur in rabbits is caused by deposition of the pigment melanin in hairs. Melanin is produced from the amino acid tyrosine through a series of chemical reactions. Production of melanin depends on the possession of the dominant allele C^B.

Which sequence of events correctly describes the link between genotype and phenotype in all black-furred rabbits?

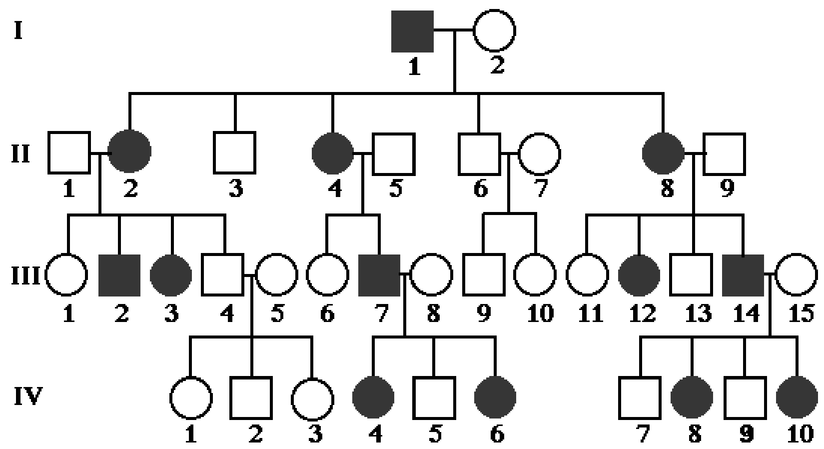
A C^B DNA replicated → C^B DNA translated at ribosomes → tyrosine polymerized into melanin

B C^B allele translated → polypeptide transcribed → folded to give functional melanin

C C^B DNA transcribed → mRNA translated → enzyme catalyses melanin production

D Two C^B alleles expressed in all cells → enzyme synthesised → melanin protein produced

22 The inheritance of a genetic disease in a family is presented in a pedigree tree below.



What is the **most** likely type of inheritance shown?

A autosomal dominant

B autosomal recessive

C sex-linked dominant

D sex-linked recessive

23 Albino horses have recessive alleles which cannot be transcribed to produce pigmentation. Geneticists studying horses have found a different allele of the same gene which predisposes carriers to premature aging of their follicles. This allele is dominant, so, even when born with pigmented hair, the coats of horses with this allele will gradually turn white over the first 6 to 8 years of life.

Which **white** female horse would be the best to use in crosses to determine the genotype of a ten-year-old, white male horse?

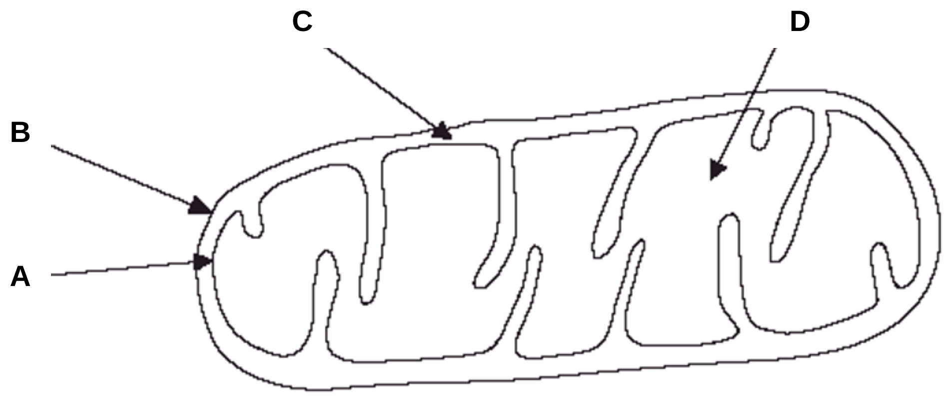
A an unrelated, ten-year-old female

B an unrelated, three-year-old female

C a sister of the male

D the male's mother

24 Where is carbon dioxide produced in the mitochondrion?



25 The formation of ethanol from pyruvate is an example of

A aerobic respiration.

B alcoholic fermentation.

C carbon dioxide production.

D lactate fermentation.

26 Which of the following stages is carbon dioxide released from catabolism?

1 glycolysis

2 oxidative phosphorylation

3 Krebs cycle

4 link reaction

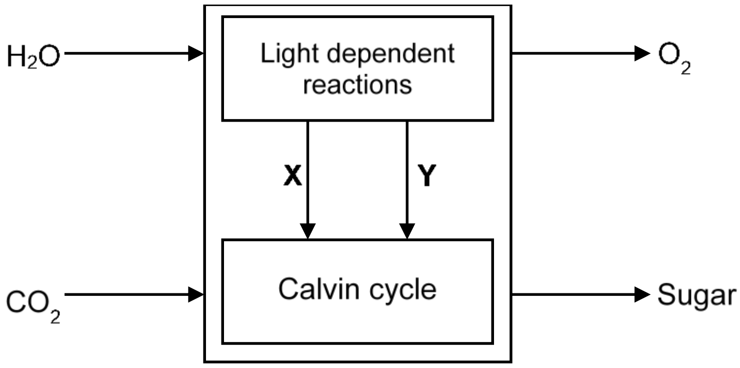
A 1 and 3 only

B 1 and 4 only

C 2 and 3 only

D 3 and 4 only

27 The diagram below shows the processes that occur in the chloroplast during photosynthesis.

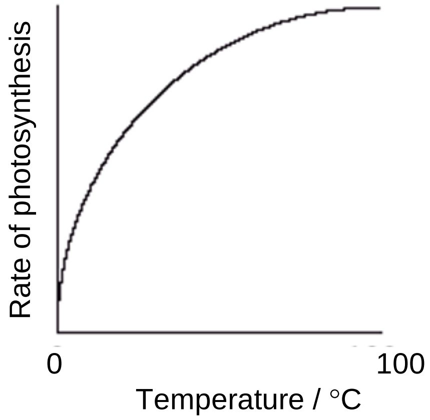


What are **X** and **Y**?

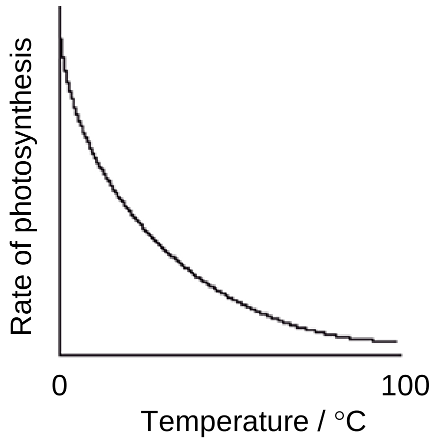
	X	Y
A	ATP	reduced NAD
B	ATP	reduced NADP
C	H^+	Electrons
D	reduced FADH	reduced NAD

28 Which graph best represents the effect of temperature on the rate of photosynthesis of a plant?

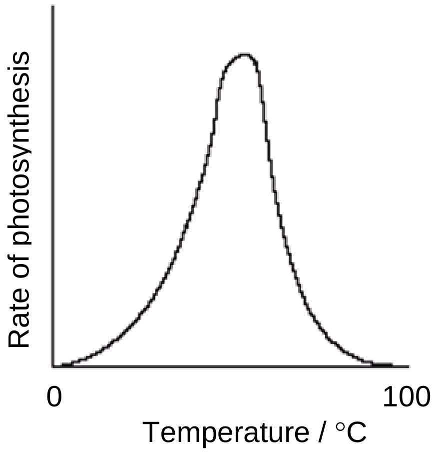
A



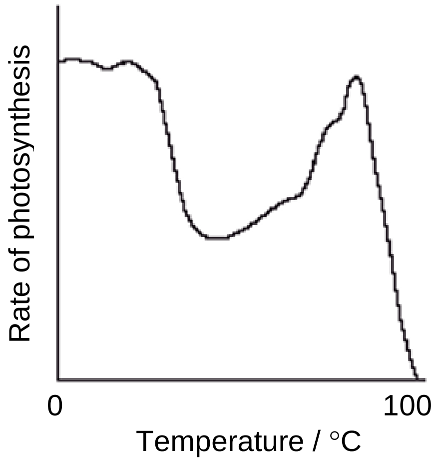
B



C

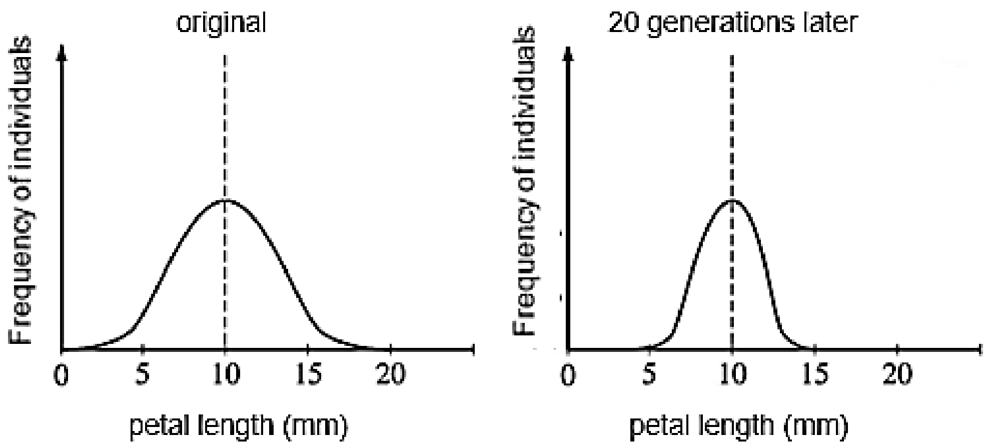


D



29 A plant species arrives at a new island and are exposed to a new set of pollinators.

The diagram below shows the frequency distribution of petal length in the original colonising population and 20 generations later.



Which type of selection is shown in this example?

A directional selection

B disruptive selection

C neutral selection

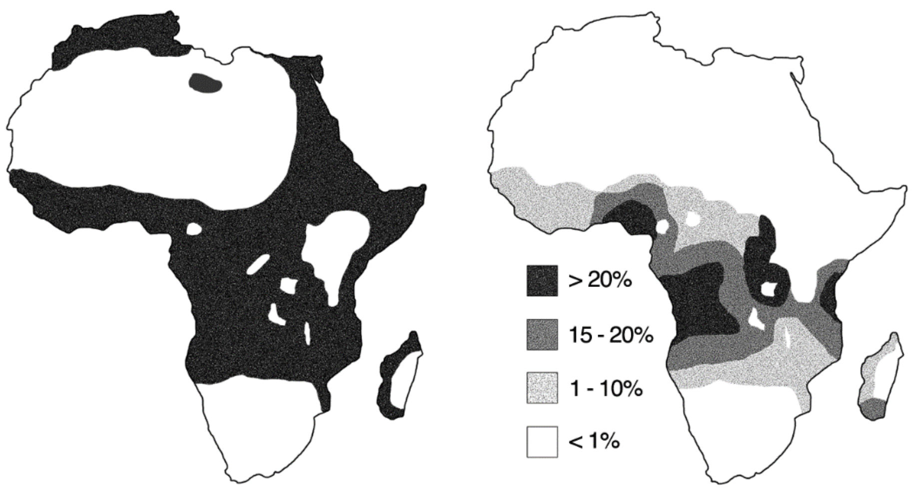
D stabilising selection

30 Sickle cell anaemia is a disease resulting in an abnormality in the haemoglobin found in red blood cells. Individuals with sickle cell anaemia are unable to transport oxygen effectively and suffer high mortalities. These individuals have two copies of the recessive sickle cell allele.

Malaria is a disease caused by a parasite that is transmitted to humans through the bites of infected *Anopheles* mosquitoes. The parasites infect red blood cells, resulting in destruction of red blood cells.

In Africa, the frequency of the sickle cell allele is high. The figure shows the distribution of malaria and the sickle cell allele in Africa.

malaria-prone areas **distribution of sickle cell allele**



Which of the following best explains the high frequency of the sickle cell allele in Africa?

A There is high incidence of the sickle cell allele in malaria-prone areas.

B The malarial parasite is unable to infect sickled red blood cells.

C Individuals with sickle cell anaemia will have a lower likelihood of having malaria.

D In heterozygotes, the sickle cell allele can be passed down to offspring.

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ANSWERS

Qn	Ans
1	A
2	C
3	C
4	B
5	C
6	A
7	D
8	D
9	A
10	B

Qn	Ans
11	C
12	B
13	D
14	C
15	C
16	B
17	C
18	A
19	C
20	B

Qn	Ans
21	C
22	C
23	B
24	D
25	B
26	D
27	B
28	C
29	D
30	D